

September 2022 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *No additional materials*

Calculators: *Calculators are NOT allowed in this examination*

1. Total: 15 points (18 minutes)

Suppose that a pair of fair dice is tossed. Let random variable X denote the sum of the two numbers obtained. That is, X can equal x , for $x = 2, 3, \dots, 12$.

- (a) Find $\Pr(X = x)$, for $x = 6$. **(3pts)**
 - (b) Work out a general expression for $\Pr(X = x)$, for any x . (Alternatively, you may list the values of the probability function over its entire support.) **(4pts)**
 - (c) Let A denote the event that “ X is divisible by 4”, and let B denote the event that “ X is greater than 9”. Find $\Pr(A)$, $\Pr(B)$, and $\Pr(A \cap B)$. **(4pts)**
 - (d) State Bayes’ theorem in terms of events A and B . Verify numerically that it holds. **(4pts)**
-

2. Total: 15 points (18 minutes)

Consider the following density function,

$$f_{X,Y}(x, y) = \begin{cases} k(x^2 + y^2), & \text{for } 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What are the conditions under which $f_{X,Y}$ is a valid density function? **(5pts)**
 - (b) Suppose you are interested in working out $\Pr(X < Y^2)$. Sketch the region (in the domain of $f_{X,Y}$) over which you would integrate $f_{X,Y}$ to find the solution. Identify axis labels, titles and label any lines/curves that you draw. **(5pts)**
 - (c) Hence, or otherwise, show that $\Pr(X < Y^2) = 26/35$. **(5pts)**
-

3. Total: 10 points (12 minutes)

Consider random variable X with probability function

$$\Pr(X = x) = \begin{cases} (1 - \pi)^x \pi, & \text{for } x = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Show that the moment-generating function of X , say $M_X(t)$, is given by

$$M_X(t) = \frac{\pi}{1 - e^t(1 - \pi)},$$

and state values of t for which $M_X(t)$ is well-defined. **(5pts)**

- (b) Use $M_X(t)$ to derive $E(X)$. **(5pts)**

4. **Total: 15 points (18 minutes)**

Consider an estimator $\hat{\theta}_N$ for an unknown parameter θ , where N denotes the sample size.

- (a) Define formally the properties of bias, variance and mean-square error of $\hat{\theta}_N$.
Derive an exact relationship between these three properties. **(5pts)**
- (b) Define formally what is meant by consistency of $\hat{\theta}_N$ (**without** using the “ p lim” operator). **(4pts)**
- (c) Suppose we have a random sample of size $N \geq 3$ given by $\{X_i\}_{i=1}^N$ from the $\text{Normal}(\mu, 1)$ distribution. Let $\theta = \mu^2$, and let $\bar{X}_N$ denote the sample mean. Your colleague suggests:

$$\begin{aligned}\hat{\mu} &= X_1 + (X_2 - X_3)^2 - 2, \text{ as an estimator for } \mu; \text{ and} \\ \hat{\theta}_N &= \bar{X}_N^2 - 1/N, \text{ as an estimator for } \theta.\end{aligned}$$

Assess whether each estimator, $\hat{\mu}$ and $\hat{\theta}_N$, is unbiased and consistent. **(6pts)**

5. **Total: 20 points (24 minutes)**

- (a) Suppose we have a set of mutually independent random variables $X_i \sim \text{Normal}(0, \sigma^2)$ for $i = 1, \dots, N$, where σ^2 is unknown.
 - i. Derive the method of moments (MM) estimator for σ^2 . **(2pts)**
 - ii. Derive the maximum likelihood (ML) estimator for σ^2 . (You do **not** need to state the second-order condition.) **(6pts)**
- (b) Now suppose we have a different set of mutually independent random variables $Y_i \sim \text{Normal}(i\mu, i^3\sigma^2)$ for $i = 1, \dots, N$, where μ is unknown but σ^2 is known.
 - i. Derive the ML estimator for μ . (You do **not** need to state the second-order condition.) **(10pts)**
 - ii. Comment on the distribution of the ML estimator. **(2pts)**

Hint: You may find it helpful to recall that if $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \text{ for any } x \in \mathbb{R}.$$

6. Total: 10 points (12 minutes)

You have several identical-looking coins in your pocket; most are fair coins, but there is one coin that comes up heads with probability $1/4$. You pick a coin at random and flip it 10 times. You decide that you will reject the hypothesis that the coin is fair if it comes up heads 3 times or fewer.

- (a) Set this up as a formal hypothesis test. What is the null hypothesis, alternative hypothesis, test statistic, and decision rule? **(4pts)**
- (b) Derive explicit algebraic expressions for the size and power of this test (but do **not** attempt to solve for their exact numerical values). **(4pts)**
- (c) Suppose that the coin comes up heads only once. Derive an explicit algebraic expression for the corresponding p -value (but do **not** attempt to solve for its exact numerical value). **(2pts)**

Hint: You may find it helpful to recall that if $X \sim \text{Binomial}(n, p)$, then

$$\Pr(X = x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

7. Total: 15 points (18 minutes)

- (a) Explain *carefully* what is meant by a $100(1 - \alpha)\%$ confidence interval (CI) for an unknown parameter μ , for some $\alpha \in (0, 1)$. **(2pts)**
- (b) Suppose we have a random sample $\{X_i\}_{i=1}^N$ from a distribution with finite mean μ and finite positive variance σ^2 . Provide a formal statement of the (Lindeberg-Lévy) central limit theorem (CLT). **(2pts)**
- (c) Suppose we have a random sample $\{X_i\}_{i=1}^N$ from a $\text{Normal}(\mu, \sigma^2)$ distribution with finite mean μ and finite positive variance σ^2 , where σ^2 is known.
 - i. Derive a $100(1 - \alpha)\%$ CI for μ **without** relying on the CLT. **(5pts)**
 - ii. How could one use the CLT to conduct inference on μ ? How does the CLT-based approach compare with the method in part (i) above? **(3pts)**
 - iii. Now assume σ^2 is unknown. Explain how to conduct inference on μ in this case. Your answer should explain whether the CLT is needed or not. **(3pts)**